Metrovale Metro Operations and Service Review

Detailed operational report | Fictional dataset for General RAG testing | FY 2025–26

Executive Summary

Metrovale Metro is a fictional urban public transportation network designed to provide a realistic, internally consistent document for retrieval-augmented generation experiments. The network operates three lines—Blue, Green, and Orange—and serves 42 stations across 68 route-kilometres. During the 2025–26 financial year, average weekday ridership reached 312,000 passenger journeys. Reported on-time performance was 92.8%, below the network's target of 95%. The results suggest that service reliability and preventive maintenance are priority areas for improvement.

This report describes network operations, passenger demand, fares, accessibility, maintenance practices, incident response, and performance improvement. Figures are synthetic and must not be treated as real-world transit data.

1. Network Structure and Operating Hours

The Blue Line, Green Line, and Orange Line form the core of the Metrovale network. Together, they cover 42 stations and 68 route-kilometres. The Blue Line has 18 stations and carries approximately 46% of all passenger journeys. The Green Line has 14 stations and accounts for 34% of journeys. The Orange Line has 10 stations and accounts for the remaining 20%. These figures imply that the Blue Line is the largest contributor to network demand.

On weekdays, scheduled passenger service begins at 05:30 and ends at 23:30. During peak periods, trains are planned to arrive every 6 minutes. During off-peak periods, the planned interval is 10 minutes. Actual intervals may vary during incidents, engineering work, or unusually high passenger demand. The Network Control Centre coordinates train movements and communicates operational changes to station teams.

When interpreting operating statistics, it is important to distinguish a planned service interval from on-time performance. A six-minute peak interval describes the intended frequency of trains, whereas on-time performance measures how consistently services meet their published schedule. Improving one measure does not automatically guarantee improvement in the other.

2. Passenger Demand and Service Reliability

Average weekday ridership for FY 2025–26 was 312,000 passenger journeys. Based on the reported line shares, the Blue Line carried approximately 143,520 journeys per weekday, the Green Line carried 106,080, and the Orange Line carried 62,400. These line-level values are calculated from the total and the stated percentages; they are not separate passenger counts from a ticketing system.

The network's on-time performance result was 92.8%, compared with a target of 95%. The gap is 2.2 percentage points. A useful investigation would compare delays by line, time of day, station dwell time, train availability, and incident category. Because the Blue Line carries the highest share of journeys, disruption on that line may affect more passengers than a similar disruption on a lower-demand line. This is a reason to prioritize analysis, not proof that the Blue Line is responsible for the network-wide reliability gap.

Service planners should review recurring delay patterns before changing train frequency. Increasing frequency can improve capacity, but it can also reduce recovery time between trains and make a congested service more sensitive to small disruptions. Operational decisions should be based on observed demand, dwell times, headway regularity, and the causes of delay.

3. Fare Policy and Passenger Information

Metrovale uses a distance-based single-journey fare. Trips up to 5 kilometres cost 2.50 credits. Trips over 5 kilometres and up to 12 kilometres cost 3.25 credits. Trips longer than 12 kilometres cost 4.00 credits. A monthly pass costs 78 credits and provides unlimited travel on all three lines. Children under 6 travel free when accompanied by a fare-paying adult.

For a 10-kilometre journey, the applicable single fare is 3.25 credits because the journey falls in the band over 5 kilometres and up to 12 kilometres. A journey of exactly 5 kilometres costs 2.50 credits. A journey of exactly 12 kilometres costs 3.25 credits, while a journey longer than 12 kilometres costs 4.00 credits. These boundary definitions are stated explicitly to reduce ambiguity when answering user questions.

Passenger information should communicate fare bands, pass eligibility, service hours, and disruption notices in clear language. If fare policy changes, the effective date and superseded rules should be recorded so that a retrieval system can distinguish current guidance from historical information. The present report describes one fictional policy snapshot for FY 2025–26.

4. Accessibility and Passenger Support

All 42 Metrovale stations are described as having step-free access from street level to the platform. Tactile paving is installed at platform edges, and passenger help points are available at every station. These provisions are intended to support passengers with mobility, visual, hearing, or other access needs, although the exact assistance required will vary by passenger and station conditions.

Elevators receive a visual inspection each day and a technical inspection every 30 days. A safety-critical defect requires the elevator to be taken out of service, signs to be placed at both entrances, and an accessible alternative route to be communicated. The target repair time is 4 hours, but the elevator must not reopen until safety clearance has been granted. The repair target is an operational objective, not permission to bypass inspection or clearance.

Station supervisors are responsible for local passenger assistance. They should communicate the status of access equipment and identify alternative routes when an elevator or other accessibility feature is unavailable. Information should be consistent across station announcements, displays, and the network's passenger communication channels.

5. Preventive Maintenance and Asset Reliability

Train inspections take place before the first scheduled departure each operating day. Standard checks include brakes, door interlocks, onboard communications, lighting, and emergency equipment. A train that fails a brake test must not enter passenger service. Non-critical defects must be recorded before the end of the shift and assigned a maintenance priority.

Track patrols are conducted every 7 days. Engineering controllers classify detected defects as critical, major, or minor. Critical defects require immediate protection of the affected section and notification to the Network Control Centre. Signal-system backups are tested on the first Sunday of each month between 00:30 and 03:30, when passenger service is closed.

The target for completion of planned preventive-maintenance tasks is 98%. The reported completion rate for FY 2025–26 was 96.5%, which is 1.5 percentage points below target. Repeat failures within 30 days were 4.2%, compared with a target below 3%. These results suggest that the operator should examine work-order closure quality, spare-part availability, inspection consistency, and root-cause analysis for recurring faults.

Maintenance metrics should be interpreted together. A high percentage of completed work orders is not sufficient if the same assets repeatedly fail soon after maintenance. Likewise, a low repeat-failure rate should not be achieved by deferring necessary work. The aim is safe asset availability supported by accurate records and effective corrective action.

6. Incident Management and Service Disruption

When a disruption is expected to exceed 10 minutes, the control centre must publish an initial passenger notice within 5 minutes of confirmation. If a line is blocked for more than 20 minutes, the Operations Duty Manager evaluates replacement buses and cross-platform transfers. The appropriate response depends on the location of the incident, available capacity, passenger demand, and whether a safe transfer route exists.

For smoke, fire, or suspected electrical arcing, staff must alert the control centre immediately, keep passengers away from the affected area, and follow the station evacuation plan. Equipment must not be re-energized until the electrical controller authorizes it. For a medical incident, the report should identify the station, platform, time, and assistance required while limiting personal medical details to what responders need.

After service recovery, an incident report is due within 24 hours. A useful report records the timeline, affected assets and services, passenger impact, communications issued, decisions made, and follow-up actions. The purpose is to support learning and reduce recurrence, rather than merely documenting that an incident occurred.

7. Roles, Responsibilities, and Escalation

The Network Control Centre coordinates train movements and network-level incident communications. Station supervisors manage local passenger assistance and report station conditions. The Operations Duty Manager approves service suspensions and evaluates replacement transport. Engineering controllers assess track and signaling defects, while the electrical controller authorizes re-energization after electrical incidents. Clear role boundaries reduce delays and help ensure that safety-critical decisions are made by the appropriate authority.

When an incident crosses team boundaries, the first recipient should pass the report to the responsible function and confirm that it has been acknowledged. Logs should record the time of escalation and the decision owner. For a serious incident, passenger communications, engineering actions, and service-control decisions should be coordinated through the designated incident process.

8. Recommended Improvement Priorities

First, investigate the 2.2-percentage-point gap between on-time performance and its 95% target. Analyze delay causes by line and time period, and distinguish primary incidents from delays propagated through the network. Second, raise preventive-maintenance completion from 96.5% toward the 98% target while reviewing why repeat failures are 4.2%, above the target of less than 3%. Third, review passenger communication timelines and verify that disruption notices meet the stated service standards.

Fourth, audit accessibility-equipment outage handling, including signage, alternative-route information, repair-time tracking, and safety clearance before reopening. Fifth, improve the quality of operational data by recording consistent incident categories, timestamps, asset identifiers, and action owners. These recommendations are based on the synthetic figures in this report and are intended as test scenarios for document retrieval, not as advice to a real operator.

Key Facts for Retrieval Testing

The network has 42 stations and 68 route-kilometres. Weekday service runs from 05:30 to 23:30. Peak train intervals are 6 minutes and off-peak intervals are 10 minutes. Average weekday ridership is 312,000 journeys. The Blue Line accounts for 46% of journeys, the Green Line 34%, and the Orange Line 20%. On-time performance is 92.8% against a 95% target. Single fares are 2.50 credits up to 5 km, 3.25 credits over 5 km and up to 12 km, and 4.00 credits over 12 km. The monthly pass costs 78 credits. The safety-critical elevator repair target is 4 hours. Preventive-maintenance completion is 96.5% against a 98% target, and repeat failures are 4.2% against a target below 3%.

Document note: Metrovale Metro and all associated operational details are fictional. This PDF contains no tables and is intended for testing PDF extraction, chunking, retrieval, and question answering in a General RAG

application.